Nexus State Warehouse

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Problem

Scientific data is fragmented:

- SAXS $\rightarrow$ structure: $p(r)$, R_g, D_{max}
- SPR $\rightarrow$ kinetics: k_a, k_d
- Cryo-EM $\rightarrow$ geometry: surface and curvature
- PCA $\rightarrow$ thermodynamics: G-space and low-dimensional state maps

Current failure mode:

- no unified query layer
- no shared state representation
- no cross-domain inference object

Vision

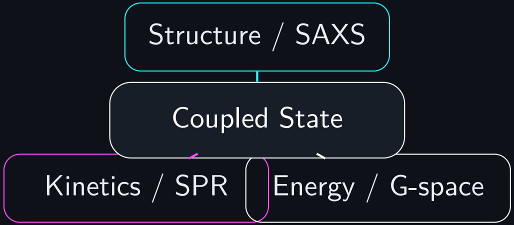
Replace datasets with **states**

State = minimal physically consistent description of a system

Instead of querying experiments, Nexus queries biological and physical states that multiple experiments constrain together.

Core Concept

- Structure → SAXS
- Geometry → Cryo-EM
- Kinetics → SPR
- Energetics → PCA / G-space



Architecture

RAW DATA → FEATURES → STATES → WAREHOUSE → HYPOTHESES

The warehouse is the middle layer where heterogeneous experimental evidence becomes one queryable state object.

Data Model

- Dataset
- Feature
- State
- Model
- Hypothesis

These are not business tables. They are scientific abstractions for linking observables to admissible physical states.

SQL Schema

```sql
CREATE TABLE datasets (
  dataset_id TEXT PRIMARY KEY,
  method TEXT,
  path TEXT,
  temperature FLOAT,
  ph FLOAT
);

CREATE TABLE features (
  feature_id TEXT PRIMARY KEY,
  dataset_id TEXT,
  name TEXT,
  value FLOAT,
  unit TEXT
);

CREATE TABLE states (
  state_id TEXT PRIMARY KEY,
  system TEXT,
  label TEXT,
  confidence FLOAT
```

Scientific Query Example

Question: Does structural expansion correlate with stronger binding?

```
SELECT s.state_id
FROM states s
JOIN state_features sf1 ON s.state_id = sf1.state_id
JOIN features f1 ON sf1.feature_id = f1.feature_id
JOIN state_features sf2 ON s.state_id = sf2.state_id
JOIN features f2 ON sf2.feature_id = f2.feature_id
WHERE f1.name = 'Rg' AND f1.value > 5.0
  AND f2.name = 'kd' AND f2.value < 0.002;
```

Hypothesis Layer

Example state:

- R_g increases
- k_d decreases

Hypothesis:

- structural expansion stabilizes binding
- confidence is derived from multi-modal agreement

Why This is Different

Classical data warehouse

- business metrics
- static tables
- dashboards

Nexus warehouse

- physical states
- cross-domain coupling
- hypothesis generation

Modern Stack

Core:

- DuckDB for local analytical execution
- Apache Arrow for columnar transport
- Python for orchestration and scientific execution

Extensions:

- graph database for state relations
- feature store for ML integration
- versioned object storage for large experimental payloads

Alternative Architectures

Graph-based:

- direct state-to-feature and state-to-state relations
- natural representation of coupled systems

Lakehouse:

- versioned datasets
- time-travel queries
- reproducible warehouse snapshots

Your Edge

AI learns correlations

Nexus learns physics

Conclusion

Nexus State Warehouse enables querying biological reality